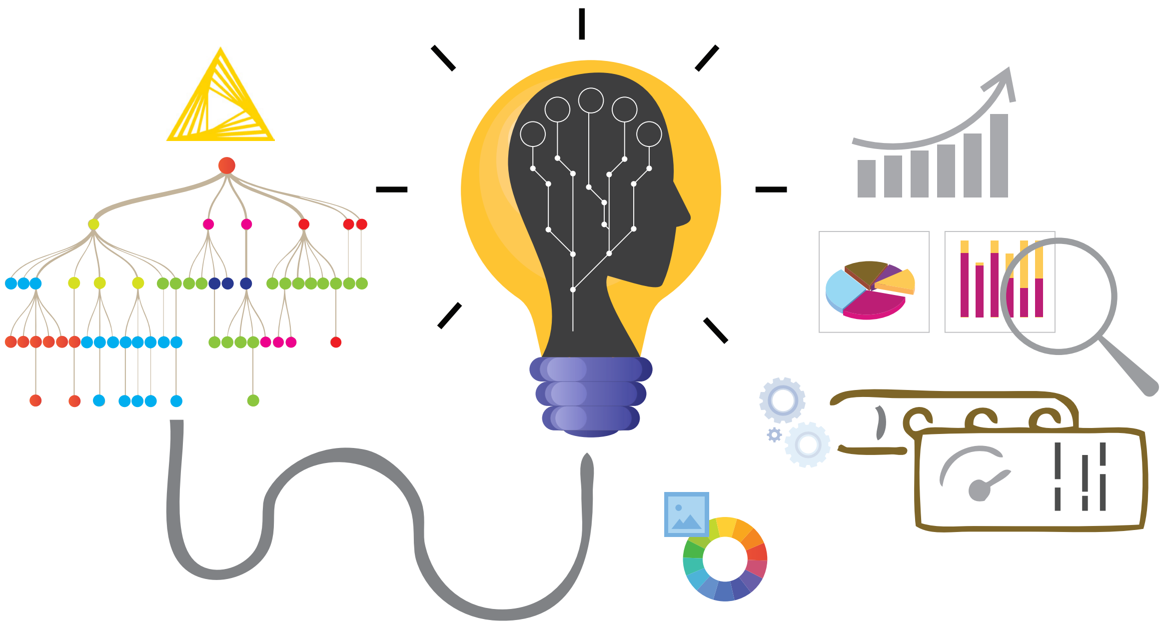
Supervised Machine Learning for Car Evaluation Using KNIME

The KNIME (Konstanz Information Miner) analytics platform is perhaps the strongest and most comprehensive free platform for drag-and-drop analytics, machine learning, statistics, and ETL (extract, transform and load) models. This article introduces a supervised machine learning decision tree algorithm through the KNIME tool for a car evaluation data set.



Data pre-processing, modelling, analysis, and visualisation are all enabled within KNIME. The workflows can run both through the interactive interface and also in batch mode. These two setups allow for easy local job management and regular process execution. KNIME integrates various components for machine learning and data mining through its modular data pipelining 'Lego of Analytics' concept.

Let us look at a supervised machine learning decision tree algorithm through the KNIME tool for the car evaluation data set. Decision trees are created for the car model based on the technical features, pricing criteria (overall cost, maintenance cost), comfort, size of luggage boot, safety, passenger capacity, etc. The decision tree learner node and the predictor nodes are connected with both trained data and test data for training and evaluating the model with better output. Then the decision tree predictor is configured with the scorer node to generate the required evaluation performance metrics and to report data for user interface visualisation.

KNIME installation

To start with, we will first install KNIME. Download the KNIME desktop application from *https://www.knime.com/* and install it as per the instruction on the screen. You will have to provide the necessary permissions during the setup process.

Decision tree

A decision tree is a decision support tool that uses a tree-like model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility. It is one way to construct a model that contains conditional control statements. Decision trees are used for handling non-linear data sets effectively. The decision tree tool is used in real life in many fields, such as engineering, civil planning, law, and business.

Figure 1 lists the nodes that we will be using to create the decision tree.